

Solving 1st-order linear ODEs with the panserformel

Panserformel = integrating factor method

The ODE family we want to solve

$$y' + p(t)y = q(t)$$

Main trick

$$\mu(t) = e^{\int p(t) dt}$$

Multiply by the right factor so that
the left side becomes $(\mu y)'$

Goal: turn the left side into one derivative, then integrate.

Why molecular biology students care

Many biology models have the same pattern:
rate of change = input - proportional loss.

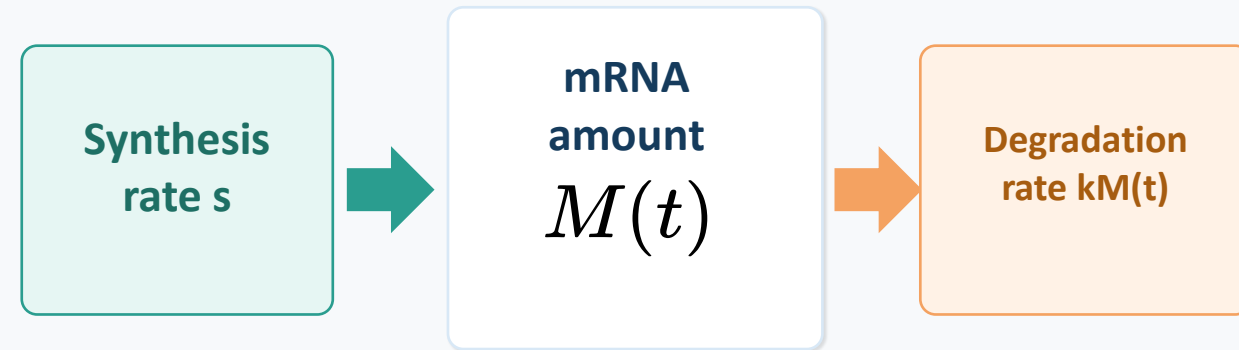
Example: let $M(t)$ be an mRNA amount.

Transcription adds molecules at rate s , while
degradation removes them at rate $kM(t)$.

$$M'(t) = s - kM(t)$$

$$M'(t) + kM(t) = s$$

This is a 1st-order linear ODE.



Same mathematical structure appears in protein turnover, signaling, and drug clearance.

In this mini-lesson you will learn how to recognize the form, understand the trick, and solve it step by step.

When separation works - and when it fails

Separation works when the equation can be written as

$$\frac{dy}{dt} = A(t)B(y)$$

Example

$$y' = ty$$
$$\frac{1}{y} dy = t dt$$

All t-terms and y-terms can be separated.

A linear ODE has the form

$$y' + p(t)y = q(t)$$

Example

$$y' + y = t$$
$$y' = t - y$$

This is usually not a product $A(t)B(y)$, so separation is not the systematic choice.

Important: some special linear ODEs can also be separated. The panserformel is still the reliable method for the whole linear family.

The idea behind the panserformel

The method uses only one new idea: choose a function $\mu(t)$ so that the left side matches the product rule exactly.

$$y' + p(t)y = q(t)$$



Multiply by $\mu(t)$

$$\mu y' + \mu p(t)y = \mu q(t)$$



Use 1) and 2) to realize that the left side can become one derivative

$$(\mu y)' = \mu q(t)$$

1) Remember the product rule

$$(\mu y)' = \mu y' + \mu' y$$

2) To make the two expressions match, choose μ so that

$$\begin{aligned}\mu'(t) &= p(t)\mu(t) \\ \mu(t) &= e^{\int p(t) dt}\end{aligned}$$

That is the whole trick: multiplying by $\mu(t)$ packages two terms into one derivative, so the ODE becomes easy to integrate.

The step-by-step recipe

$$y(t) = e^{-P(t)} \left(\int e^{P(t)} q(t) dt + C \right)$$

Here $P'(t)=p(t)$, so $P(t)$ is any antiderivative of $p(t)$.

- 1 Rewrite the ODE in standard form $y' + p(t)y = q(t)$.
- 2 Find the integrating factor $\mu(t) = e^{\int p(t)dt}$.
- 3 Multiply the entire equation by $\mu(t)$.
- 4 Rewrite the left side as $(\mu y)'$.
- 5 Integrate, solve for y , and use the initial condition.

Important sign check

The integrating factor uses $+\int p(t)dt$.

The minus sign appears only later, when you divide by $\mu(t)$.

Best habit: memorize the 5 steps, not only the final formula.

Worked example: set up the method

Solve $M'(t) + M(t) = 2$ with $M(0) = 0$.

Interpretation: constant synthesis 2 and proportional degradation $1 \cdot M(t)$.

1 Standard form $M' + 1 \cdot M = 2 \quad \Rightarrow \quad p(t) = 1, q(t) = 2$

2 Integrating factor $\mu(t) = e^{\int 1 dt} = e^t$

3 Multiply the ODE $e^t M' + e^t M = 2e^t$

4 Product derivative $\frac{d}{dt}(e^t M) = 2e^t$

Why step 3 becomes step 4

$$\frac{d}{dt}(e^t M) = e^t M' + e^t M$$

So the left side really is a product derivative.

This example is chosen because the arithmetic is gentle, so the method is easy to see.

Worked example: finish the solution

Continue
from

$$\frac{d}{dt}(e^t M) = 2e^t$$

5 Integrate both sides

$$e^t M = \int 2e^t dt = 2e^t + C$$

6 Solve for M(t)

$$M(t) = 2 + Ce^{-t}$$

7 Use the initial
condition

$$0 = M(0) = 2 + C \Rightarrow C = -2$$

Final answer

$$M(t) = 2 - 2e^{-t}$$

Memory line

standard form

→ integrating factor

→ product derivative

→ integrate

→ initial condition

Check and interpret the answer

Quick check

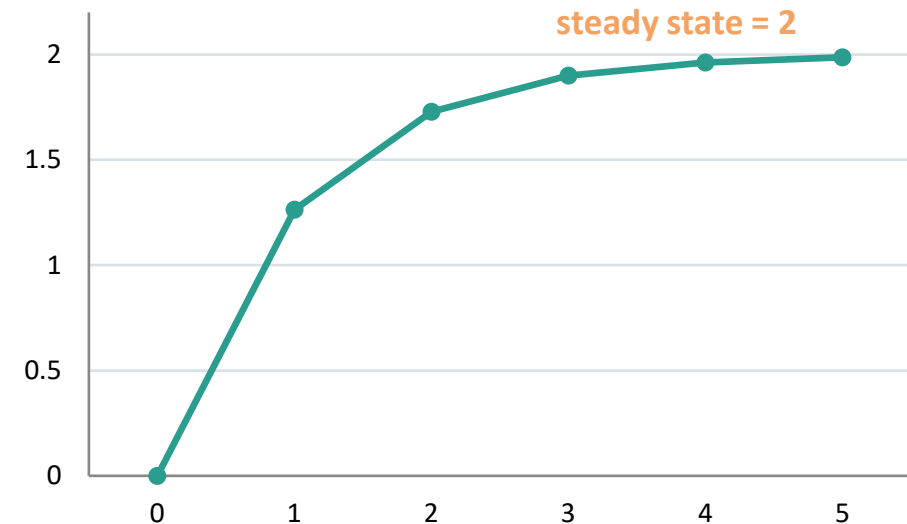
$$M'(t) = 2e^{-t}$$

$$M'(t) + M(t) = 2e^{-t} + 2 - 2e^{-t} = 2$$

$$M(0) = 2 - 2 = 0$$

So both the differential equation and the initial condition are satisfied.

What the solution does



starts at 0
rises quickly
levels off near 2

Biological meaning: as M grows, degradation catches up with constant synthesis, so the net change becomes smaller and smaller.

Self-study checklist

Before you start any new problem

- 1 Can I rewrite it as $y' + p(t)y = q(t)$?
- 2 What are $p(t)$ and $q(t)$?
- 3 What is $\mu(t)=e^{\int p(t)dt}$?
- 4 After multiplying, can I see $(\mu y)'$?
- 5 Did I use the initial condition and check the answer?

Most common mistakes

- forgetting to put the ODE in standard form first
- using the wrong sign in $\mu(t)$
- multiplying only one term by $\mu(t)$
- stopping before using the initial condition

Try this yourself: $y' + 2y = 6$, $y(0)=1$ Answer: $y(t)=3-2e^{-2t}$